

Problems in Two dimensional Random Variables

1)

If X denotes the number of aces and Y the number of queens obtained when 2 cards are drawn at random (without replacement) from a deck of cards, obtain the joint probability distribution of (X, Y) .

2)

The joint probability function of two discrete RVs X and Y is given by $f(x, y) = c(2x + y)$, where x and y can assume all integers such that $0 \leq x \leq 2$ and $0 \leq y \leq 3$, and $f(x, y) = 0$ otherwise. (a) find the value of c and (b) find $P(X \geq 1, Y \leq 2)$. (MKU — Apr. 96)

3)

The input to a binary communication system, denoted by a RV X , takes one of two values 0 or 1 with probabilities $3/4$ and $1/4$ respectively. Because of errors caused by noise in the system, the output Y differs from the input occasionally. The behaviour of the communication system is modeled by the conditional probabilities given below:

$$P(Y = 1/X = 1) = 3/4 \text{ and } P(Y = 0/X = 0) = 7/8$$

Find (i) $P(Y = 1)$, (ii) $P(Y = 0)$ and (iii) $P(X = 1/Y = 1)$.

4)

The following table represents the joint probability distribution of the discrete RV (X, Y) . Find all the marginal and conditional distributions.

Y	X		
	1	2	3
1	1/12	1/6	0
2	0	1/9	1/5
3	1/18	1/4	2/15

5)

If the joint pdf of a two-dimensional RV (X, Y) is given by

$$f(x, y) = x^2 + \frac{xy}{3}; 0 < x < 1, 0 < y < 2$$

$$= 0, \text{ elsewhere}$$

find (i) $P\left(X > \frac{1}{2}\right)$, (ii) $P(Y < X)$ and (iii) $P\left(Y < \frac{1}{2} / X < \frac{1}{2}\right)$.

6)

If the joint pdf of a two-dimensional RV (X, Y) is given by

$$f(x, y) = k(6 - x - y); 0 < x < 2, 2 < y < 4$$

$$= 0, \text{ elsewhere}$$

find (i) the value of k , (ii) $P(X < 1, Y < 3)$,
(iii) $P(X + Y < 3)$ and (iv) $P(X < 1/Y < 3)$.

7)

Given that the joint pdf of (X, Y) is

$$f(x, y) = e^{-y}; x > 0, y > x$$

$$= 0, \text{ elsewhere}$$

find (i) $P(X > 1/Y < 5)$ and (ii) the marginal distributions of X and Y .

8)

If the joint pdf of the RV (X, Y) is given by $f(x, y) = k(x^3y + xy^3)$, $0 \leq x \leq 2$, $0 \leq y \leq 2$, find (i) the value of k , (ii) the marginal densities of X and Y and (iii) the conditional densities of X and Y .

9)

The joint density function of a RV (X, Y) is $f(x, y) = 8xy$, $0 < x < 1$, $0 < y < x$. find the conditional density function $f(y/x)$. (MU — Apr. 96)

10)

The two-dimensional RV (X, Y) has the joint density

$$f(x, y) = 8xy, 0 < x < y < 1$$

$$= 0, \text{ otherwise}$$

- (i) Find $P(X < 1/2 \cap Y < 1/4)$,
- (ii) Find the marginal and conditional distributions, and
- (iii) Are X and Y independent? Give reasons for your answer.

Answer:

1)

$\begin{array}{c} Y \backslash X \\ \hline \end{array}$	0	1	2
0	0.71	0.13	0.01
1	0.13	0.01	0
2	0.01	0	0

2)

$$c = \frac{1}{42}; P = \frac{4}{7}$$

3)

(i) $\frac{9}{32}$

(ii) $\frac{23}{32}$

(iii) $\frac{2}{3}$

4)

$\{i, p_{i*}\}$	$\{j, p_{*i}\}$	CPD of $X/Y = 1$	CPD of $X/Y = 2$																																
<table><tr><th>$x = i$</th><th>p_{i*}</th></tr><tr><td>1</td><td>5/36</td></tr><tr><td>2</td><td>19/36</td></tr><tr><td>3</td><td>1/3</td></tr></table>	$x = i$	p_{i*}	1	5/36	2	19/36	3	1/3	<table><tr><th>$y = j$</th><th>p_{*j}</th></tr><tr><td>1</td><td>1/4</td></tr><tr><td>2</td><td>14/45</td></tr><tr><td>3</td><td>79/180</td></tr></table>	$y = j$	p_{*j}	1	1/4	2	14/45	3	79/180	<table><tr><th>$x = i$</th><th>p_{i1}/p_{*1}</th></tr><tr><td>1</td><td>1/3</td></tr><tr><td>2</td><td>2/3</td></tr><tr><td>3</td><td>0</td></tr></table>	$x = i$	p_{i1}/p_{*1}	1	1/3	2	2/3	3	0	<table><tr><th>$x = i$</th><th>p_{i2}/p_{*2}</th></tr><tr><td>1</td><td>0</td></tr><tr><td>2</td><td>5/14</td></tr><tr><td>3</td><td>9/14</td></tr></table>	$x = i$	p_{i2}/p_{*2}	1	0	2	5/14	3	9/14
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5)

(i) $\frac{5}{6}$

(ii) $\frac{7}{24}$

(iii) $\frac{5}{32}$

6)

(i) $1/8$

(ii) $3/8$

(iii) $5/24$

(iv) $3/5$

7)

$$\frac{31}{256}$$

8)

(i) $\frac{e^4 - 5}{e^5 - 6}$

(ii) $f_x(x) = e^{-x}, x > 0; f_y(y) = y e^{-y}, y > 0.$

9)

$$K = 1/16; f_x(x) = \frac{1}{8} (x^3 + 2x), 0 \leq x \leq 2; f_y(y) = \frac{1}{8} (y^3 + 2y), 0 \leq y \leq 2;$$

$$f(x/y) = x/2 \frac{(x^2 + y^2)}{(y^2 + 2)} \quad 0 < x < 2; f(y/x) = y/2 \frac{(x^2 + y^2)}{(x^2 + 2)}, 0 < y < 2$$

10)

$$f(y/x) = 2y/x^2, 0 < y < x$$

11)

(i) $1/256$

(ii) $f_x(x) = 4x (1 - x^2)$ in $(0, 1)$; $f_y(y) = 4 y^3$ in $(0, 1)$; $f(y/x)$

$$= \left| \frac{2y}{1 - x^2} \right| f(x/y) = \frac{2x}{y^2}$$

(iii) X and Y are not independent.